EVERY DG MODULE ADMITS A SURJECTION FROM A DG MODULE WITH FINITELY GENERATED FREE QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-good-quotient (source lines 2489-2502); the proof environment begins two lines below it and is its own proof, closing at source line 2516.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the admissible sequence and the shifts are the author's own surrounding development in the same chapter.

Editorial note. The statement's three clauses are displayed in an enumerate environment and are reproduced verbatim; no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: every dg module receives a surjection from a dg module whose cycles surject and whose two halves are direct sums of shifts.

Allowed standard prerequisites (editorial).

- Differential graded module over a dg algebra. A graded A-module with a differential d of degree one satisfying the Leibniz rule, so that its cycles form a graded submodule.. It is the ambient category, and the two surjectivity clauses are statements about M and its cycles (source lines 2491-2500, 2512-2515).
- Admissible short exact sequence of dg modules. A short exact sequence of dg modules which splits as graded modules, so that the middle term is the direct sum of the outer terms.. Clause (3) asks precisely for such a sequence with the outer terms direct sums of shifts of A, which is what the construction of P provides (source lines 2498-2500, 2510-2512).
- Shifted free module A[k]. The free dg A-module on one generator concentrated in degree k, with trivial differential.. Adding direct sums of A[k] is how clause (2) is achieved without spoiling clause (3) (source lines 2512-2515).

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules with the following properties*

(1) *$P \to M$ is surjective,*

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(2) $\mathrm{Ker}(d_P) \to \mathrm{Ker}(d_M)$ *is surjective, and*
(3) P *sits in an admissible short exact sequence* $0 \to P' \to P \to P'' \to 0$ *where* P', P'' *are direct sums of shifts of* A.

Proof. Let P_k be the free A-module with generators x, y in degrees k and $k+1$. Define the structure of a differential graded A-module on P_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. For every element $m \in M^k$ there is a homomorphism $P_k \to M$ sending x to m and y to $\mathrm{d}(m)$. Thus we see that there is a surjection from a direct sum of copies of P_k to M. This clearly produces $P \to M$ having properties (1) and (3). To obtain property (2) note that if $m \in \mathrm{Ker}(\mathrm{d}_M)$ has degree k, then there is a map $A[k] \to M$ mapping 1 to m. Hence we can achieve (2) by adding a direct sum of copies of shifts of A. $\square$